

RESEARCH STUDY ON ETHICAL DILEMMAS IN CROSSOVER'S ARTIFICIAL INTELLIGENCE RECRUITMENT SOFTWARE

Introduction

Software systems that are powered by Artificial intelligence (AI) are being implemented at a record-breaking pace in society and disrupting several industries (Rashid, 2024). We are finding that AI is being used even in settings and cases which previously we could never have dreamt of (Lee, 2020). The recruitment industry has not been exempted from the proliferation of AI systems. The traditional job application and hiring process which is established based on personal qualities, in-person assessment days/ interviews and body language observation is gradually being erased by AI powered recruitment solutions. In 2024, the UK government's Department for Science, Innovation and Technology (DSIT) took notice of this trend of AI recruiters and published its official guidance on the responsible deployment of AI systems for Human Resources and Recruitment professionals (UK Dept for Science, Innovation and Technology (DSIT), 2024). This guidance was published after the UK Government engaged with HR and Recruitment organisations to understand their attitudes towards trustworthy AI.

The findings from this engagement demonstrated that there is need for recruitment professionals to define clearly what role AI software is supposed to play in their organisation to avoid potential misuse. According to DSIT, some of the suggested questions that recruiters and their contractors who design the AI systems are expected to ask before deploying AI systems are:

- A) "Does introducing technology into the recruitment process create new barriers to applicants with protected characteristics?
- B) Does introducing technology into the recruitment process amplify existing risks at scale, for example, entrenching and/or perpetuating human biases?
- C) Does introducing technology into the recruitment process create novel risks at scale?" (UK Dept for Science, Innovation and Technology, 2024)

This study was designed to capture ordinary people's responses to the above questions despite none of them being recruiters, job applicants at Crossover/its subsidiaries or civil servants with any of the government's departments. None are professional or aspiring artificial engineers with foreknowledge of what ethical and trustworthy AI frameworks exist so their perception of Crossover's AI chatbot is not biased. Majority of them agreed that the disadvantages to vulnerable group using this system outweighed the advantages. They agreed unanimously that Crossover benefited from this platform.

I will empirically analyse the answers those interviewed have given to identify themes and patterns. I will also ask GenAI the same questions to analyse how its responses differs and match with human responses.

Research Question and Objectives

To gather anonymised and unbiased data from a random sample of lay people (with no Artificial Intelligence training) regarding the perceived impact and ethical implications of a radically different AI recruitment process.

The core research questions that interviewees will be asked to gauge their perceptions are:

- 1) What do you think of an automated application process that gives you no opportunity to submit a CV or Cover Letter but is merely a Chatbot?
- 2) Who do you think benefits from an AI chat recruiter and how do they benefit?
- 3) Who do you think is at a disadvantage from an AI chat recruiter and how are they at a disadvantage?

Significance of this study in the UK's Recruitment Eco-system

The decision to deploy AI technologies as a filtering and autonomous decision-making system in the recruitment process has proven to be a sound business decision for organisations for several reasons. For example:

1) Recruitment affects an organisations overall performance (Chanavatalda, 2015) and the right staff is seen as pivotal in a business's overall growth. Therefore, a standardised procedure managed by an AI algorithm/predictive model provides a benchmark which all applicants need to pass through safeguarding the company's quality of personnel. The company's "barriers to entry" and quality of employee performance do not vary based on who the interviewing human resources manager is since software will decide based only the criteria explicitly given.

2) Many industries in the UK are experiencing a long-term skills shortage such as manufacturing, healthcare, construction and engineering. The green/net zero industry is particularly affected because there is a demand for 400,000 jobs which is almost double the current supply for 216,000 jobs (Institute of Environmental Management and Assessment "IEMA", 2025). This will negatively impact the UK's net zero target which is set for 2050. Businesses that have a shortage of managerial employees and experts report Brexit as a factor in their skills shortages (Tiwasing, 2020). The timing of Brexit coincided with the rapid growth of AI usage in UK industries with McKinsey reporting that 56% of companies were using AI in at least one business function in 2021 which is an increase from 50% in 2020 (UK Parliament, House of Commons). This means that as the skills shortage continues to impact

businesses, AI will continue to become increasingly more entrenched in several industries' processes (including human resources and recruiting cycles).

3) Most online jobs have far more applicants than recruiters to look through them and even when there are enough recruiters to look through them, there are not enough work hours in a week to manually look through applications especially when applications can amount to thousands for just one vacancy in certain positions. Therefore, there is high demand for software tools that will automate and increase efficiencies in screening applications. Software also tests applicants that have passed screening.

4) Any organisation that is averse to AI will inevitably fall behind its rivals that adopt AI. (King, 2019). Industries that are interested in getting and keeping their competitive edge must not fall behind those who are becoming increasingly more efficient with custom predictive models that assist them in a highly competitive business environment.

Ethical Considerations & The Burden of Avoiding Potential Misuse

To balance out the clear benefits above of using AI recruitment systems, ethical considerations and trade-offs being made by prototypical users of AI (i.e. Developers, Business owners and impacted groups) must take centre stage. As subtly summarised by Hervieux; "It is now clear that AI applications are not neutral technological solutions. They can embellish and magnify prejudices and stereotypes, and they can perpetuate errors and limitations". (Hervieux, 2022).

Hervieux highlights the implicit bias problem between humans and data by pointing out that human coders are a key source of bias. These human biases in turn imbue biased algorithms and are embedded into AI algorithms since subconscious and conscious biases accompany everything that humans do. It is yet to be established whether Crossover's engineers have used their software to filter out any applicants whose background and skills do not match current and previous employees which is against a diverse workforce with diverse paradigms/backgrounds. Diverse teams have been proven to be smarter and more productive (Rock, 2016).

Additionally, Crossover's AI chatbot is not exempt from the Law of Unintended Consequences since if we are to filter out everyone who doesn't meet AI's specific measurements, then we are also to remove current Crossover employees who don't meet strict AI criteria for job performance benchmarks at job reviews. When AI decided who gets hired, should AI not also decide who remains in their current role?

GDPR 2016 and Explainable AI

Article 22 of GDPR has stated that there are legal limits to decision making based solely on automated processing and profiling. People also have the right to meaningful explanations

whenever they are trying to understand the automated system that they have encountered to make sure that it doesn't break the law as stated in Article 22. Therefore, Crossover has the legal burden of ensuring that AI is merely a tool in hiring and not the final judge of who gets hires. Some of those interviewed for the study below ultimately echoed the same sentiments in their own words.

Method

Members of the public who explicitly consented to be interviewed in person were shown a video recording of a brand-new applicant going through Crossover's chatbot recruiter. It was a real job application for an application that was still live for the position of a senior artificial intelligence engineer. Once they had watched the video to their satisfaction and asked all questions to better understand why no cv or cover letter could be attached they were all asked the same research questions above in an interview style.

Interview Responses

Female Therapist, British, in her 30s with a Degree in Marketing

She felt that an automated application process is new technology. She was anxious of what will be asked by the chatbot and what its decision making is based on. She asked if the chatbot is essentially allowing candidates repeat what is normally sent in a CV and I told her that after using the one hour allocated to respond to all questions, it asked for less than 20% of what will normally be contained on a CV. She mentioned that this can be repetitive if you were applying for various jobs with Crossover's subsidiaries who also use the same chatbot.

Benefits and Disadvantages:

She felt that the organisations benefit because not much money or staff resources are allocated for interviews. The applicants maybe do not really benefit as they might spend one hour on this process without receiving feedback on their performance. Maybe applicants could be at an advantage if they have social anxiety with traditional small talk which is eliminated here so socially anxious people may do well. Those that are not socially anxious get less of an advantage due to less human and company connection building. Both parties will not really know what each person is like as there is no body language that can be observed. There is no way to know if you like the person being interviewed or dislike them. With a public facing job or a sales job people's likability is essential.

Female mental health nurse, British, in her 30s. Has no degree.

Was surprised to see that AI could be used this way in such an important human activity as interviewing for a real, high paid job with a real, global business as Crossover with a large workforce and operating budget.

Benefits and Disadvantages:

She felt that neurotypical people will be the ones to benefit from Crossover's AI platform because the question and answers process online is straightforward/uncomplicated. Communication is a big thing in interviews which some neurotypical people might struggle with. Being in front of a chatbot does not appear to judge applicants based on fluency of communication. She mentioned that the platform puts them under some pressure regardless since the time limit of one hour might not be enough for certain mental health conditions.

She felt that the chance to upload a cover letter is a missed opportunity for applicants to personally connect with the jobs. Demonstrate their motivations for wanting to work for this organisation or within this department is important.

Male Administrator, British, in his 20s with two degrees; one in Philosophy and another in Political Science.

He initially questioned which applicant gets access to the chatbot and when this is done in the recruitment cycle. I told him that as soon as the applicant clicks apply under the job advert, they get a link in their email to this chatbot replacing the traditional online applicant form.

Benefits and Disadvantages:

He felt that automated interviews meant that each applicant is starting on a blank canvas so this platform is a leveller of people unlike when applicants are required to submit CVs that may make them look more desirable on face value based on their sales ability. He mentioned that the filtering may help employers save time and resources in the recruitment process. A face to face or video interview will help you read body language, voice intonation, presentation and mannerisms, etc.

He mentioned that since this is not a face-to-face interview, there is no way to verify who is really typing and that's a real issue.

He also believed that a benefit is that the employer saves time and money as they don't have to spend on HR. However, the employer is still at a disadvantage because they cannot verify that it is really the applicant who is answering questions, nor can they see if the applicant is using Generative AI to respond which is a real risk. An applicant can cheat easily in this interview and give answers that don't reflect their genuine work.

Female Home Student, British, second year of her Undergraduate degree studying illustration, 18 years.

Felt comfortable and familiar with the chatbot after watching it being used and asked no background questions.

Benefits and Disadvantages:

The company benefits because they cut costs on interviewing people. People who are at a disadvantage are those that are unfamiliar with AI tools like chatbots who might struggle with timing their answers. Also, individuals who don't know much about the job who will really want to ask many questions from an interviewer, or a HR recruiter have no opportunity to do so.

Regarding prejudice, it depends as if there's no malfunction or ulterior motive which the platform is serving behind the scenes through AI. Felt this solution will help to stop racial discrimination as there's no prejudice which the programmers or the company could quietly accomplish.

Male International Student with undergraduate in Civil Engineering. Early 20s.

Supports the use of a chatbot in recruitment. Asked several questions to understand what is wrong with a traditional job application process.

Benefits and Disadvantages:

Felt there is a strong case for using these software applications in this context. He felt that the candidates benefit from the AI chatbot as it gives them a chance to apply and be interviewed without delay as normal application processes. He concluded by saying that autonomous applications and AI chatbots should never replace face to face applications but could be used to assist anyone who will benefit from the assistance of the software.

Thematic Analysis of Responses

a) Absence of Cover Letters & CV

80% of those interviewed were surprised that a chatbot could replace a Cover Letter and CV for a real job.

b) Crossover's Overall Gains from AI

100% of those interviewed agree that Crossover and its employees benefit from this chatbot and the AI filtering system.

c) Applicants Benefit from the Chatbot

60% of those interviewed agreed that the applicant materially benefits from the AI chatbot since those that are neurodiverse and may be discriminated against in a traditional hiring process are unlikely to be discriminated against by an Artificial intelligence chatbot.

d) Applicants are at a Disadvantage from the Chatbot

40% of those interviewed believed that the applicant may be at a disadvantage from an AI chatbot.

e) AI should not Replace the Traditional Interview/Recruitment Process

60% of those interviewed felt that AI should never replace the traditional interview process.

f) Explainability of the Algorithms Decision Making Process.

40% of those interviewed questioned the decision-making process behind the chatbot determining who proceeds through to the job offer stage.

g) AI is Prejudiced to Applicants

40% of those interviewed expressed concerns of how prejudiced AI will be to applicants although only 20% felt AI is inevitably prejudiced to the applicants due to human bias of the engineers and possibly the organisation's ulterior motives.

h) AI Recruitment Chatbots Permits Cheating

20% of those interviewed strongly felt that the chatbot permits cheating as GenAI can be used to answer or another individual could respond on behalf of the applicant unlike in traditional in-person/ telephone/ video interviews.

Interviewing GenAI

I decided to gather feedback from Gemini and ChatGPT to see what their answers will be to the same interview questions answered above. Here were the responses attached below:

GEMINI

The use of AI in recruitment, particularly platforms like the one you described from Crossover, presents a fascinating and complex set of advantages and disadvantages. It's a significant departure from traditional hiring methods, and it's important to analyze its implications from various perspectives.

Here are my thoughts on your questions:

(A) What do you think of an automated application process that gives you no opportunity to submit a CV or Cover Letter but is a chatbot?

This approach represents a radical shift from conventional recruitment.

Potential Advantages:

- **Efficiency and Speed:** For high-volume roles, processing applications through a chatbot is significantly faster than manual review of hundreds or thousands of CVs and cover letters. This can drastically reduce time-to-hire.
- **Standardization:** Every applicant goes through the exact same set of questions, ensuring a consistent and standardized initial screening process. This can reduce human error or variability in initial assessments.
- **Focus on Direct Answers:** By removing CVs/cover letters, the system forces applicants to directly answer questions relevant to the role, potentially cutting through generic or embellished self-descriptions.
- **Reduced Initial Human Bias (Potentially):** If the chatbot's questions and scoring are well-designed and regularly audited, it could theoretically reduce unconscious biases that humans might introduce when reviewing traditional application documents (e.g., biases related to university names, previous company prestige, or even gender/ethnic sounding names).

Potential Disadvantages:

- **Lack of Holistic View:** CVs and cover letters, despite their flaws, provide a narrative of an applicant's career journey, accomplishments, and motivations. A chatbot, by its nature, struggles to capture this holistic context. It's hard to convey passion, unique experiences, or a compelling personal story through a series of short, directed answers.

(B) AI Chatbot vs. Human Filtering

"Better" is subjective; it's a different approach with distinct trade-offs.

Where AI Chatbot Might Be "Better" (for the company, in specific contexts):

- **Scalability:** Manages high application volumes that overwhelm human recruiters.
- **Consistency:** Applies uniform criteria to all applicants.
- **Speed:** Provides instant initial feedback, accelerating the recruitment cycle.

Where a Human Being Is Likely "Better":

- **Nuance & Context:** Humans excel at understanding complex career paths and identifying transferable skills.
- **Soft Skills & Culture Fit:** Recruiters gauge communication, enthusiasm, and cultural alignment through interaction.
- **Two-Way Communication:** Applicants can ask crucial questions about the role/company, fostering engagement and allowing them to assess fit.
- **Empathy & Candidate Experience:** Human interaction feels more valued and respectful; automated processes can be impersonal.
- **Problem-Solving/Clarification:** Humans can ask clarifying questions for ambiguous backgrounds; chatbots are typically rigid.
- **Rapport Building:** Essential for attracting top talent.

Conclusion for (B): AI chatbots offer efficiency and consistency for initial, high-volume, rule-based filtering. However, for holistic assessment, nuance, soft skills, and vital two-way communication, human recruiters remain superior. A hybrid approach, combining AI screening with human assessment, is often ideal.

(C) Beneficiaries of AI Chatbot Recruiting Software

1. The Company/Employer:

- **Cost & Time Savings:** Reduces recruiter overhead and shortens application review time.

- **Scalability:** Handles large application volumes without proportional staff increases.
- **Consistent Screening:** Aims for more objective shortlists by applying uniform criteria.
- **Data Collection:** Gathers data to refine hiring criteria and improve AI performance.
- **Wider Reach:** Easier application process may attract a broader candidate pool.

2. The AI Platform Provider (e.g., Crossover):

- **Business Model:** Generates revenue by selling services.
- **Data for Improvement:** Interactions provide data for AI model refinement.
- **Market Share:** Establishes leadership in a growing HR tech market.

3. Certain Applicants (in specific scenarios):

- **Structured Question Experts:** Those whose skills align perfectly with chatbot questions and who are adept at concise answers may progress quickly.
- **Remote Candidates:** Facilitates easier application for global roles.

(D) Disadvantages of AI Chatbot Recruiter

1. The Majority of Applicants:

- **Lack of Personal Connection:** Impersonal process can be frustrating, missing opportunities for rapport and company insight.
- **Limited Potential Showcase:** Chatbot's structured questions restrict storytelling and showcasing unique skills/growth.
- **Bias Reinforcement:** AI trained on biased historical data can perpetuate or amplify existing biases.
- **Stress & Anxiety:** Timed, impersonal format can be highly stressful.
- **Digital Literacy Disadvantage:** Less tech-familiar individuals may struggle.
- **No Feedback:** Unsuccessful candidates receive no personalized reasons for rejection.
- **Identity Verification Risks:** Lack of verification raises concerns about integrity and fairness.

CHATGPT

ChatGPT ▾

Great questions! Let's go through them one by one with some careful reasoning and analysis.

(A) What do you think of an automated application process that gives you no opportunity to submit a CV or Cover Letter but is a chatbot?

My thinking:

CVs and cover letters traditionally provide context that goes beyond yes/no or multiple-choice answers: they tell a story about your career, values, and aspirations. Removing them could result in a more standardized process, but it might also eliminate nuance and self-expression.

On the other hand, a chatbot-based process may be easier for some people (less time to write a tailored CV/cover letter), and it might reduce unconscious bias if designed properly. However, it also removes the chance for applicants to showcase their unique story and contextualize their experience, which is often crucial—especially in creative or leadership roles.

My conclusion:

It feels too restrictive and mechanical, and it risks missing out on good candidates who might not shine in a purely chatbot Q&A format. The complete removal of cover letters and CVs eliminates a valuable layer of context.

(B) Do you feel an AI chatbot is a better way to filter out candidates in comparison to a human being that you can send your CV to and ask questions before application?

My thinking:

Humans can understand nuance, adapt to the conversation, and offer clarification—something that chatbots struggle with. Humans can also answer questions applicants might have about the job (like expectations, growth opportunities), which is important for mutual fit.

However, an AI chatbot can process large volumes [↓] quickly, ensuring consistency and removing some

However, an AI chatbot can process large volumes quickly, ensuring consistency and removing some biases (e.g., it won't be influenced by a candidate's accent or appearance).

My conclusion:

A chatbot might be more efficient for high-volume, low-complexity roles, but it lacks the empathy, nuance, and adaptability of a human. For roles requiring cultural fit, creativity, or complex problem-solving, a human is better.

(C) Who do you think benefits from an AI chatbot recruiting software and how do they benefit?

My thinking:

- **Employers:** They save time and money. They can quickly filter large numbers of candidates without having to read thousands of CVs.
- **Candidates with non-traditional backgrounds:** If the chatbot is designed to focus on skills rather than traditional qualifications, it might offer a more level playing field for those who don't have standard CVs.

My conclusion:

Mainly, it's employers who benefit most directly: faster screening, lower costs, and potential standardization of evaluation criteria.

(D) Who do you think is at a disadvantage from an AI chatbot recruiter and how do they benefit?

My thinking:

This seems like there's a typo in the question (someone at a disadvantage and also benefits?), but I'll answer as: *who is at a disadvantage but might still get some small benefit.*

- **Candidates who need human interaction to**  **gain gaps, career pivots, or special circumstances:**
They lose out because chatbots don't handle nuance well.
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- **Candidates who need human interaction to explain gaps, career pivots, or special circumstances:** They lose out because chatbots don't handle nuance well.
- **But they might still benefit from a transparent scoring system (if the chatbot is well designed)** because they know exactly what's evaluated.

(E) Who do you think is at a disadvantage from an AI chatbot recruiter and how are they at a disadvantage?

My thinking:

- **Candidates from non-traditional backgrounds:** If the chatbot's questions are designed with a narrow set of assumptions, these candidates might not get the chance to show their potential.
- **Candidates with disabilities or tech issues:** They may struggle with chatbot interactions, especially under time pressure.
- **People who are better at telling their story in conversation than through short answers:** They're disadvantaged because the chatbot might not capture their strengths.

My conclusion:

Ultimately, anyone whose strengths or unique stories don't fit into the chatbot's format is at risk. This includes career changers, those with employment gaps, or those who might not be as quick at typing as they are at speaking or problem-solving.

💡 Would you like me to format this into a more formal document or do you want to expand on any of these points?

Comparison between GenAI's Interview Responses and Humans Interviewed

GenAI was able to find more examples of groups of people who are at a disadvantage from a Chatbot such as those who cannot give good answers in short quick sentences but want to tell a story to a human interviewer. Gen AI came to the same conclusion as humans that it was the organisation who benefitted from Chatbots. GenAI was able to point out that humans understand nuanced responses unlike AI chatbots that cannot pick up on nuance. This is a point that neither me nor the interviewees thought of.

GenAI identified more groups of people that will be at a disadvantage than those humans identified namely those who are from non-traditional backgrounds. However, GenAI was not able to appreciate that there is an issue with racism and discrimination in the traditional interview process which a Chatbot would most likely eliminate. The racism against cultural minorities occurs in terms of being desirable as employees in Western countries (Adamovic, 2022).

Conclusion

This research paper has demonstrated that AI recruitment software solutions are an essential tool to 21st century organisations due to multiple reasons such as high number of applicants and time constraints in manually screening. An organisation that doesn't want to be left behind by others implementing AI systems must catch up with its competitors quickly who are gaining business efficiency and competitiveness with AI tools. However, an organisation that is bound by GDPR cannot completely rely on AI for its final decision based on Art 22 or the GDPR 2016 Act. Despite the benefits to the organisation and potentially benefits to the applicant, the law of unintended consequences has demonstrated here that

implementing AI systems can have the undesirable effect of eliminating candidates that the organisation needs to thrive e.g. those who do not fit into the perfect box of the algorithm's filtering (due to their untraditional background or their status as ethnic minorities or as neurodivergent candidates) but are highly qualified for the job.

As more organisations implement AI solutions for the first time or use AI to automate a new process/workflow society is facing more unintended consequences that can easily be overlooked/minimised at the software design stage. It is time for action to be taken by the government and organisations alike before it is too late to minimise and ideally rectify all issues.

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